Non-destructive investigation by 'Ressuage'

Introduction : Investigation of the samples using this technique allows one to characterize and analyse the defects present in the sample without damaging it. Operations on these samples are conducted by using specific products which maybe colored or fluorescent, developers, cleaning products and also by using a fluorescent lamp.

Description of the products and material :

Cleaning/washing products : SKC-S/acetone

Penetrating product : Fluxo P125 (final observation conducted in daylight)

ZL 60 (fluorescent product observed with UV lamp)

Developper : Fluxo R175

Excitation of fluorescent product : UV-Lamp

5 samples made of steel are provided. You have to study any 2 samples out of the 5 samples as per your choice. The samples are numbered on the flat surface.

Some important information before starting the experiment:

Wear your lab-coat and gloves before starting the experiment

Time required for the product to penetrate varies between 5 to 20 mins.

The samples can be washed with water with a final cleaning consisting of a tissue dipped in SKC-S (if necessary)

The developer is applied very delicately to make an extremely thin layer.

The minimum delay necessary before observation is 1-3 mins.

Procedure :

- **Wash the samples with water (or with acetone in case they were dirty). Dry them properly with a tissue paper.**
- **On one sample you spray FluxoP125 and on the other sample you spray ZL60. You coat all the faces of the sample with the product. During the spraying step, hold it in the wash basin.**
- **Place the sample vertically in the plastic basin and allow the sample to rest for 5 or 10 mins without disturbing it.**
- **Next wash the sample again delicately with water. Dry it.**

- **Then, extremely delicately spray the samples with the developer. For spraying delicately keep a distance between your sample and the spray.**

- **Allow the developer to penetrate inside. For doing this keep the sample vertically in the basin for a time period of 5 mins.**

- **Observe the sample for defects using daylight for sample sprayed with Fluxo 125 and uv-lamp for sample sprayed with ZL60.**

- **Analyse the defects. You can take photos of the sample and put them in your expt. Report.**

Q1. For every sample determine the type of defect, the position of the defect and its dimension. You can either draw a schematic or put a photo of the sample pointing out the defects with the help of arrows

Q2. Determine in the two cases which method is better and why ?

Q3. Comment on the different type of defects which you observed and explain them in brief.